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It g is a standard Gaussian r.v.

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Lemma

If X is subgaussian then $\mathrm{E}X = 0$ and

$$\mathrm{P}(|X| \geq t) \leq 2\exp(-t^2/2).$$

Proof. Note $1 + aX \leq \exp aX$, so $1 + a\mathrm{E}X \leq a^2/2$ and $\mathrm{E}X = 0$. Also $\mathrm{P}(X \geq t) \leq \mathrm{E}\exp(aX - at) \leq \exp(a^2/2 - at)$.

First properties

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$$\mathrm{E}\exp aX = 1 + \sum_{n\geq 2} \frac{a^n}{n!}$$

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Important example: A convex combination of subgaussians is subgaussian. Need not be independent. This is proved by Holder's inequality.

Comparison of Gaussians and subgaussians

Notation: γ_N standard Gaussian measure on $\mathbb{R}^N$: G a $\mathbb{R}^N$-valued r.v. of law γ_N i.e. $G = (g_i)_{i \leq N}$ where $(g_i)_{i \leq}$ are standard Gaussian independent.

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The only known proof requires the full strength of the generic chaining.

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Theorem (A)

We have $\gamma_2(T, d) \leq L \mathrm{E} \sup_{t \in T}(t, G)$

Theorem (B)

If X *is subgaussian* $\mathrm{E} \sup_{t \in T}(t, X) \leq L \gamma_2(T, d)$.

Theorem

Consider $T \subset \mathbb{R}^N$, T finite and $t_0 \in T$. Let $\Gamma = \gamma_2(T, d)$. There exists a sequence $(u_k)_{k \geq 2} \in \mathbb{R}^N$ such that

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The sequence (t_n) is a "chain" linking t to t_0

end of proof

$$t - t_0 = \sum_{n \geq 0} t_{n+1} - t_n$$

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$$t - t_0 = \sum_{n\geq 0} t_{n+1} - t_n = \sum_{n\geq 0} 2^{n/2} d(t_{n+1}, t_n) \times \frac{t_{n+1} - t_n}{2^{n/2} d(t_{n+1}, t_n)}.$$

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Reminder: $\mathrm{P}((u,X) \geq v) \leq \exp(-v^2/2\|u\|^2)$

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For v larger than $L\Gamma$ the sum in the r.h.s is small.

Complements of convex sets

Lemma

Let C convex symmetric ($C = -C$) with $\gamma_n(C) \geq 3/4$. Define

$$N(x) = \inf\{\rho > 0; x \in \rho C\}.$$

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Theorem

If $\gamma_n(C) \geq 3/4$ then the complement of LC is covered by a sequence of half spaces so that the sum of their measures is $\leq 1/2$

Convex sets and subgaussian measures

Theorem

There exists a universal constant L with the following property. Assume that X is a subgaussian r.v. and consider a convex set D with $\gamma_n(D) \geq 7/8$. Then $\mathrm{P}(X \in LD) \geq 1/2$.

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$\mathrm{P}(X \in H_k) \leq \exp(-v^2 \log k/L)$. For v universal constant the sum of these probabilities is $\leq 1/2$. So $\mathrm{P}(X \notin LvC) \leq 1/2$.

Van Handel comparison theorem.

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There exists a universal constant L such that if X is subgaussian then $\mathrm{E}\varphi(X/L) \leq \mathrm{E}\varphi(G)$.

We had proved this when $\varphi(x) = N(x)$, a norm, but this is far more general.

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This is a convex function.

Consider $\varepsilon > 0$ and the convex set $D = \{\psi \leq \mathrm{E}\varphi(G) + \varepsilon\}$. By LLN, for large M we have $\gamma_{NM}(D) \geq 7/8$.

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This is the key step in proving that X is the sum of three Gaussian r.v.s, which has spectacular applications.